

BRIAN JUMA

AI Engineer | Software Engineer | Machine Learning Enthusiast

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PROFESSIONAL SUMMARY

Final-year Software Engineering student with hands-on experience in Artificial Intelligence, Machine Learning, Computer Vision, and intelligent software systems. Skilled in Python, Pandas, NumPy, Scikit-learn, TensorFlow, OpenCV, Git, and cloud technologies. Experienced in developing AI-powered applications including facial recognition systems, crop disease detection, anomaly detection, and predictive analytics. Passionate about building practical AI solutions and continuously learning modern AI engineering practices such as LLMs and Generative AI.

TECHNICAL SKILLS

Programming: Python, JavaScript, C++, PHP, SQL, React, CSS, HTML

AI & Machine Learning: Machine Learning, Deep Learning, Computer Vision, NLP, Model Evaluation, Data Preprocessing, Feature Engineering, EDA

Libraries: Pandas, NumPy, Scikit-learn, TensorFlow, Keras, OpenCV, Matplotlib

AI Tools: Jupyter Notebook, Google Colab, Git, GitHub, VS Code

Cloud & Databases: AWS (Basic), Azure (Basic), MySQL, PostgreSQL

Soft Skills: Problem Solving, Documentation, Team Collaboration, Leadership, Communication

EXPERIENCE

AI & Software Engineer | SalamaFarm (2024 - Present)

- Built AI-powered crop disease detection systems.
- Trained and evaluated machine learning models.
- Processed agricultural datasets.
- Developed REST APIs for AI model integration.
- Worked with Git for version control.
- Documented AI experiments and workflows.

PROJECTS

AI-Powered Crop Disease Detection

- Developed an AI-powered crop monitoring system using Computer Vision and sensor fusion.
- Built machine learning models for disease detection and recommendation generation.
- Created a complete AI pipeline from image capture to prediction.
- Technologies: Python, TensorFlow, OpenCV, ESP32-CAM.

Facial Recognition KYC System

- Developed a real-time facial recognition system for secure KYC verification and identity authentication.
- Implemented face detection, feature extraction, and recognition using Python and OpenCV.
- Improved recognition accuracy through image preprocessing techniques.
- Technologies: Python, OpenCV, NumPy.

Customer Churn Prediction

- Built an ML classification model to predict customer churn.
- Performed EDA, feature engineering, and model evaluation.
- Evaluated models using Accuracy, Precision, Recall, F1-score, and ROC-AUC.

Real-Time Anomaly Detection

- Designed an intelligent anomaly detection system for monitoring real-time data streams.
- Applied statistical and ML approaches to identify abnormal behavior.

EDUCATION

BSc Software Engineering | Zetech University (2023 - Expected November 2026)

Coursework: Software Engineering, Data Structures, Algorithms, Database Systems, Web Technologies, OOP, Data Science, Mobile Development.

CERTIFICATIONS

- Artificial Intelligence Training - ICT Authority Kenya & Pathways Technologies (2026)
- Power Learn Project - AI in Software Engineering
- Power Learn Project - Software Development
- Data Analytics (SPSS & STATA)